

STATCON RMS CONTROL PANEL

Power Apps Implementation Document

Statcon Electronics India Pvt. Ltd.

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1. Project Overview

The Statcon RMS Control Panel is a Microsoft Power Apps application built to manage and track RMS (Remote Monitoring System) modules across PCU (Power Conditioning Unit) assets at Statcon Electronics India Pvt. Ltd.

The app is connected to a Microsoft SharePoint List (PCU_Directory) as its backend database and handles two primary workflows:

- Assign New RMS — Register a brand new RMS entry against a PCU asset
- Reconfigure Module — Reassign an existing RMS from one PCU to another

2. Why Power Apps? (Not Excel)

Before building this solution, the team considered using Microsoft Excel as the data management tool. However, Power Apps + Microsoft Lists was chosen for the following reasons:

Factor	Excel	Power Apps + Microsoft Lists
Multi-user access	Conflicts when multiple users edit simultaneously	Multiple users can submit data at the same time safely
Data integrity	Anyone can accidentally delete/edit any row	Controlled forms prevent accidental edits
Mobile access	Limited, not user-friendly on phone	Full mobile app experience via Power Apps mobile
Validation	Manual, error-prone	Built-in required fields, dropdowns, choice columns
Workflow control	Not possible without VBA	Radio buttons control which form shows — no coding needed
Audit trail	No history unless manually maintained	SharePoint Lists automatically track who changed what and when
Scalability	Slows down with large data	SharePoint Lists handle thousands of records efficiently
Permissions	Hard to restrict who edits what	Role-based access — view only or edit per user
Integration	Manual export needed	Directly integrates with Teams, Power Automate

3. Business Benefits

3.1 Operational Benefits

- Eliminates manual entry errors through dropdown choices and form validation
- Real-time data — all team members see the latest information instantly
- Full history of RMS assignments and reconfigurations maintained automatically
- STATUS tracking (Active, Recycle, Pending, Reconfigured) gives clear visibility of asset state

3.2 Management Benefits

- No IT dependency — app built and maintained by the team using no-code Power Apps
- Accessible from anywhere — office, field, or mobile

- Scalable — new PCU models or columns can be added anytime without rebuilding the app
- Audit ready — SharePoint automatically logs every change with timestamp and user name

3.3 Cost Benefits

- Zero additional software cost — Power Apps and SharePoint are included in existing Microsoft 365 license
- No developer needed — built entirely using no-code/low-code tools
- Reduces time spent on manual tracking in spreadsheets

4. Technology Stack

Component	Technology	Purpose
Frontend App	Microsoft Power Apps	UI, forms, logic, navigation
Database	Microsoft Lists (SharePoint)	Data storage for PCU directory
Backend List	PCU_Directory	Main data source for all operations
Hosting	SharePoint (statconelectronics)	Cloud hosting via Microsoft 365
Deployment	Power Apps Publish & Share	Distribute to team members
Integration	Power Automate (future scope)	Automated notifications & workflows

5. Microsoft List Structure (PCU_Directory)

Column Name	Data Type	Purpose
Dealer/Company	Single line of text	Name of the dealer or company
PCU serial number	Single line of text	Unique serial number of the PCU
Specifications/Rating	Choice	PCU rating — 31 options from 2kVA/48V to 150kVA/480V
Project No./Workorder	Single line of text	Associated project or work order number
STATUS	Choice	Asset state: Active, Recycle, Pending, Reconfigured
RMS ID	Single line of text	Unique RMS module identifier

Column Name	Data Type	Purpose
Previous PCU	Single line of text	PCU serial no. for which RMS was configured earlier

6. App Features

6.1 Radio Button Workflow Selector

A radio button control (radWorkflowType) at the top switches between two modes. All form elements show or hide based on this selection using Visible property formulas.

Radio Option	Visible Controls	Action
Assign New RMS	Form7 + Submit New Entry button	Creates a new record in PCU_Directory
Reconfigure Module	Form_Reconfigure + cmbOldPCU dropdown + Execute button	Updates an existing PCU record

6.2 Assign New RMS

Allows users to register a new RMS entry. Connected to PCU_Directory via Form1 using SubmitForm for automatic data submission.

- Fields: Dealer/Company, PCU serial number, Specifications/Rating, Project No./Workorder, STATUS, RMS ID
- Submit Button OnSelect formula:

```
SubmitForm(Form1); Notify("Entry Submitted Successfully!", NotificationType.Success)
```
- Visibility formula:

```
radWorkflowType.Selected.Value = "Assign New RMS"
```

6.3 Reconfigure Module

Allows users to reassign an existing RMS from one PCU to another. The dropdown shows only PCUs available for reconfiguration (i.e. STATUS = Recycle).

- Dropdown (cmbOldPCU) Items formula:

```
Filter(PCU_Directory, STATUS.Value = "Recycle")
```
- Previous PCU auto-fill (Default property):

```
cmbOldPCU.Selected.'PCU serial number'
```
- Execute Reconfiguration Button (OnSelect):

```
Patch(PCU_Directory, cmbOldPCU.Selected,
{
    'Previous PCU': cmbOldPCU.Selected.'PCU serial number',
    'PCU serial number': DataCardValue.Text,
    'Specifications/Rating': txtNewSpecs.Text,
    STATUS: {Value: "Pending"}
})
```

```
});  
Notify("Reconfiguration Successful!", NotificationType.Success)
```

- Visibility formula:

```
radWorkflowType.Selected.Value = "Reconfigure Module"
```

6.4 PCU Specifications Dropdown

The Specifications/Rating field is a Choice column with 31 predefined options covering all PCU models manufactured by Statcon:

Range	Options Available
Small (2-6 kVA)	2kVA/48V, 3kVA/48V, 3.5kVA/24V, 5kVA/48V, 5kVA/96V, 5kVA/240V, 5.5kVA/48V, 6kVA/48V
Medium (7.5-20 kVA)	7.5kVA/96V, 10kVA/48V, 10kVA/120V, 10kVA/240V, 12.5kVA/120V, 15kVA/120V, 15kVA/240V, 20kVA/120V, 20kVA/240V
Large (25-100 kVA)	25kVA/240V, 30kVA/240V, 40kVA/240V, 50kVA/240V, 50kVA/360V, 60kVA/360V, 80kVA/360V, 100kVA/240V, 100kVA/360V
Extra Large (120-150 kVA)	120kVA/360V, 120kVA/480V, 125kVA/360V, 125kVA/480V, 150kVA/480V

7. Implementation Steps

Step	Action	Details
1	Created Microsoft List	PCU_Directory with all required columns in SharePoint
2	Connected List to Power Apps	Add data → SharePoint → PCU_Directory
3	Added Radio Button	radWorkflowType with Assign New RMS / Reconfigure Module options
4	Built Assign New RMS Form	Form1 connected to PCU_Directory, all fields added
5	Set Visibility Logic	Forms show/hide based on radio button selection using Visible property
6	Built Reconfigure Section	cmbOldPCU dropdown + text inputs + Execute Reconfiguration button
7	Added Previous PCU Column	New column in PCU_Directory to track reconfiguration history
8	Linked Dropdown to Field	Auto-fill Previous PCU from dropdown selection using Default property

Step	Action	Details
9	Configured Patch Formula	Execute button updates existing record with new PCU data
10	Added Specifications Choices	31 PCU models added to Specifications/Rating choice column
11	Published & Shared App	Published via Power Apps Studio, shared with team members

8. Key Formulas Reference

Control	Property	Formula
Form1	Visible	radWorkflowType.Selected.Value = "Assign New RMS"
Submit Button	Visible	radWorkflowType.Selected.Value = "Assign New RMS"
Form_Reconfigure	Visible	radWorkflowType.Selected.Value = "Reconfigure Module"
cmbOldPCU	Items	Filter(PCU_Directory, STATUS.Value = "Recycle")
cmbOldPCU	Value	"PCU serial number"
Previous PCU DataCard	Default	cmbOldPCU.Selected.'PCU serial number'
Submit New Entry	OnSelect	SubmitForm(Form1)
Execute Reconfiguration	OnSelect	Patch(PCU_Directory, cmbOldPCU.Selected, {...})
Specifications Dropdown	Items	Choices(PCU_Directory.'Specifications/Rating')

9. Issues Faced & Solutions

Issue	Root Cause	Solution
Form showing on both radio options	Visible property not set	Set Visible = radWorkflowType.Selected.Value = "Assign New RMS"
Columns missing after list update	Power Apps caches schema	Refresh data source via Add data panel
Red squiggles in formula	Wrong column names used	Used Ctrl+Space autocomplete to get exact column names
Dropdown showing nothing	STATUS column name mismatch	Used STATUS.Value = "Recycle" without quotes on column name

Issue	Root Cause	Solution
Previous PCU not auto-filling	Default property not set on DataCard	Set Default = cmbOldPCU.Selected.'PCU serial number'
Form fields showing as custom	Form connected to wrong data source	Changed DataSource to PCU_Directory and re-added all fields
Dealer/Company field missing	Field not added to form	Edit Fields → + Add field → Dealer/Company
Specifications showing Item1/Item2	Column type was not set to Choice	Changed column type to Choice and added all 31 options

10. Deployment

10.1 Publishing

- Click Save (Ctrl+S) in Power Apps Studio
- Click Publish → Publish this version
- Wait for confirmation message

10.2 Sharing

- Click Share button in Power Apps Studio
- Search for team members by name or email
- Also grant Edit access to PCU_Directory SharePoint list

10.3 Accessing the App

- Web: make.powerapps.com → Apps section
- Mobile: Power Apps mobile app (iOS / Android)
- Teams: Add as a tab in Microsoft Teams channel

11. Future Scope

- Power Automate integration — send email notifications on new RMS assignments
- Power BI dashboard — visualize RMS utilization and PCU status across locations
- Barcode/QR scanning — scan PCU serial numbers instead of manual entry
- Approval workflow — manager approval before reconfiguration is executed
- Edit & View screens — browse existing records and edit them from the app